

Rushal Butala

Software Engineer • AI / Product Systems
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Technical Skills

Languages: TypeScript, Python, Java, C#, C++, Go (learning), SQL

AI / Data: Vector Databases (RAG), LLM orchestration, data modeling, real-time pipelines, ETL

Backend / Cloud: FastAPI, ASP.NET Core, Node.js, PostgreSQL, MongoDB, GCP, AWS, Azure, CI/CD

Frontend / Tools: React, Angular, Docker, Git, GraphQL, Figma

CS Fundamentals: Algorithms, data structures, concurrency, caching, performance optimization

Experience

Borough — Founding Engineer (Jul 2025 – Nov 2025)

- Architected and shipped full-stack infrastructure for real-time communication between residents and local governments.
- Built FastAPI + PostgreSQL services with async operations and indexing, reducing latency by 25%.
- Integrated vector-DB-powered RAG retrieval pipelines for contextual responses.
- Deployed and scaled services on Google Cloud Platform with CI/CD pipelines.

Savance — Software Engineer Intern (Jan 2024 – May 2024)

- Developed backend ETL pipelines processing high-volume logistics data.
- Built optimized TypeScript APIs improving data throughput by 30%.
- Designed normalized SQL schemas supporting reliable reporting workflows.

University of Michigan — Graduate Researcher (2023 – 2024)

- Engineered a high-availability vector search system for low-latency LLM-guided debugging.
- Reduced retrieval latency by 45% through async orchestration and parallel lookup.
- Co-authored an academic paper on optimization-aware LLM systems.

Education

University of Michigan — MS Information (HCI + Systems), 2025

University of Illinois at Urbana-Champaign — BS Computer Science, 2023